STOCK MARKET ANALYSIS USING SQL

Project Insights Report

Project Title: Stock Market Analysis Using SQL
Database: MySQL
Project Type: Data Analysis
Stocks Analysed: Bajaj Auto, Eicher Motors, Hero MotoCorp, Infosys, TCS, and TVS Motors

1. Introduction

Stock market analysis helps us understand the historical performance of stocks, identify price trends, and study changes in stock prices over time. SQL (Structured Query Language) is a powerful tool for storing, managing, retrieving, and analysing structured data.

In this project, historical stock market data for six companies was analysed using MySQL. SQL queries were used to examine closing prices, calculate moving averages, generate BUY/SELL/HOLD signals, calculate daily returns, identify significant price drops, and study bonus-related price adjustments.

2. Objectives of the Project

The main objectives of this project were:

- To understand and analyse historical stock market data using SQL.
- To retrieve the earliest and latest trading dates for a stock.
- To identify the highest closing prices of Eicher Motors.
- To calculate yearly average closing prices of TCS.
- To identify records containing missing deliverable quantities.
- To calculate 20-day and 50-day moving averages for Bajaj Auto.
- To combine data from six stock tables into a master table.
- To generate BUY, SELL, and HOLD signals using moving averages.
- To calculate daily stock returns and identify major daily price falls.
- To study bonus-related price changes in Infosys and TCS.

3. Dataset Overview

The project uses historical stock market data for six companies:

1. Bajaj Auto
2. Eicher Motors
3. Hero MotoCorp
4. Infosys
5. TCS

6. TVS Motors

The dataset covers the period from 1 January 2015 to 31 July 2018.

Each stock table contains 888 records. The six tables were combined into a master table containing 5,328 records.

Important columns include Date, Open Price, High Price, Low Price, Close Price, Deliverable Quantity, and other trading-related attributes.

4. Tools and Technologies Used

MySQL: Used to create tables, retrieve records, filter data, perform calculations, and analyse stock prices.

SQL: Used for aggregate functions, date conversion, joins, UNION ALL, window functions, moving averages, conditional statements, and data updates.

Microsoft Word: Used to prepare the project insights report.

5. Analysis Performed

5.1 Bajaj Auto Date Analysis

The Bajaj Auto table was examined to determine its total number of records, earliest trading date, and latest trading date. The table contained 888 records, covering the period from 1 January 2015 to 31 July 2018.

5.2 Highest Closing Prices of Eicher Motors

The closing prices of Eicher Motors were sorted in descending order to identify the five highest closing-price records. This analysis helps identify periods when the stock traded at relatively high closing prices.

5.3 Yearly Average Closing Price of TCS

The average closing price of TCS was calculated for each year. This provides a year-wise overview of the stock's historical price levels and helps compare its performance across different years.

5.4 Missing Deliverable Quantity

The six stock tables were checked for records with missing deliverable quantities. This analysis helps identify incomplete data that may require further investigation before performing detailed analysis.

5.5 Moving Average Analysis

The 20-day and 50-day moving averages were calculated for Bajaj Auto.

A moving average smooths short-term price fluctuations and helps identify the general direction of a stock's price trend. The 20-day moving average reacts more quickly to recent changes, while the 50-day moving average represents a longer period.

5.6 Master Table Creation

The records of all six stocks were combined into a single master table using UNION ALL.

The resulting master table contained 5,328 records, making it easier to perform combined analysis across all six companies.

5.7 BUY, SELL, and HOLD Signals

Moving averages were used to classify stock records into three signal categories:

- **BUY:** The 20-day moving average was greater than the 50-day moving average.
- **SELL:** The 20-day moving average was lower than the 50-day moving average.
- **HOLD:** Both moving averages were equal.

These are simplified, rule-based classifications for learning and analysis. They are not guaranteed trading recommendations.

5.8 Daily Returns Analysis

Daily returns were calculated by comparing each day's closing price with the previous trading day's closing price.

This analysis helps measure daily price changes and identify periods of positive or negative performance.

5.9 Identification of Major Daily Price Falls

The largest daily percentage declines identified in the dataset were:

Stock	Date	Daily Price Change
TCS	31 May 2018	-50.40%
Infosys	15 June 2015	-49.86%
TVS Motors	3 May 2016	-9.66%
Eicher Motors	24 August 2015	-9.33%
Bajaj Auto	24 August 2015	-9.09%
Hero MotoCorp	23 July 2018	-6.20%

The unusually large declines in TCS and Infosys occurred around their respective bonus-related price changes. Therefore, these movements should not automatically be interpreted as ordinary market losses. Corporate actions and adjustment methods must be considered when analysing historical prices.

5.10 Bonus Adjustment Analysis

Separate copies of the Infosys and TCS tables were created to preserve the original datasets.

A 1:1 bonus adjustment was applied to selected historical price columns before the relevant bonus dates in the copied tables. This adjustment was intended to make historical prices more comparable across the bonus events.

The original stock tables were retained separately. Adjusted data should be checked for consistency before being used in further calculations.

6. Key Findings

- Six stock tables were analysed using MySQL.
- Each original stock table contained 888 records.
- The combined master table contained 5,328 records.
- Moving averages were used to study stock-price trends.
- BUY, SELL, and HOLD classifications were generated using a moving-average comparison.
- Daily returns helped identify large changes in closing prices.
- TCS and Infosys showed unusually large price declines around bonus-related events.
- Separate adjusted tables were created to preserve the original data.
- SQL made it possible to perform repeatable analysis on multiple stock datasets.

7. Limitations

- The analysis is based on historical data and does not predict future stock prices.
- Moving-average signals are simplified indicators and may not account for all market conditions.
- Large price changes may be affected by corporate actions, data quality issues, or other factors.
- The analysis does not include all factors that influence stock prices, such as company financial performance, market sentiment, or economic conditions.
- The results should not be treated as financial advice.

8. Conclusion

This project demonstrated how SQL can be used for stock market data analysis. By using filtering, aggregate functions, date conversion, combined tables, window functions, moving averages, and conditional logic, meaningful information was extracted from historical stock data.

The project helped develop practical skills in database management, data cleaning, analytical query writing, trend analysis, and interpretation of stock-price movements. It also highlighted the importance of considering corporate actions when comparing historical stock prices.

Overall, the project provided practical experience in using MySQL to transform raw stock market data into useful analytical insights.